

A Sealed Negative for Schwinger Scaling in a Resolved-Field Hidden-Dynamics Family

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A local hidden dynamics observed through a fold in its projected time can produce apparent pair events. Whether it can also produce the field dependence of the Schwinger exponential, without that law being inserted by hand, is the sharpest question a campaign of this kind can ask. This paper reports the answer for one declared family, under seal, and the answer is no. In a resolved Sauter-slab energy-transfer model the suppression of time reversal is a Gaussian measure tail in a probe-measured effective gap, and a Schwinger-shaped alternative fails to compete on held-out data. The paper reports the full chain that produced this verdict. A four-iteration pilot turned each of its failures into a design constraint. A first sealed run returned the verdict neither-axis because its bars, set in binomial units, tested a precision claim nobody had made, and that verdict is preserved rather than repaired. An owner-imposed correction rule, declared in writing before any second attempt existed, required the next preregistration either to model the two-variable structure of the exponent explicitly or to set its bars against model error. The second sealed run took the second branch and passed both bars with room. The effective-gap model predicted gaps it had never seen better than it fit the gaps it trained on, and the Schwinger-shaped model was more than three times worse on the same held-out cells. The result licenses no statement about other families, nothing about physical pair production, and nothing about quantum electrodynamics. The question it sharpens is whether any constructible hidden-dynamics family has an exponent that is not a measure tail.

I. INTRODUCTION

An earlier paper of this campaign proved that a smooth trajectory on a hidden manifold, observed through a projection into spacetime, produces an apparent particle-antiparticle pair whenever the observer's time functional has a nondegenerate critical point along the trajectory [8]. That statement is a theorem about kinematics and it is cheap. The expensive question is dynamical. Can a specified local hidden dynamics, with no target law inserted, produce reversal statistics with the field dependence of the Schwinger pair-production exponential [1–4]? The campaign's design documents call this the Schwinger scaling challenge, and they gate it behind two contracts.

The first contract is anti-circularity. A candidate model may not contain the Schwinger exponent as an input, may not weight or select its initial conditions by anything equivalent to the target law, may not postselect on trajectories that reverse, and may not calibrate on the parameter range used for evaluation. The second contract is the closure-prescription clause. A companion paper of this campaign replicated a published classical pair-creation event [6] and proved that its reversal threshold sits exactly on the pole of the Cayley transform that the published impulsive closure applies to the interaction, while continuous accumulation of the same interaction provably never reverses [9]. A discretization can manufacture a threshold, complete with a physically appealing energy condition, that the dynamics it approximates does not have. The clause therefore requires continuous integration through every interaction, a preregistered integration prescription, agreement between two independent prescriptions at every

claim-bearing point, and stability under refinement.

This paper reports the first sealed passage through that gate. The declared family is a resolved Sauter-slab model in which a hidden thermal oscillator can transfer momentum to the projected time coordinate while the event crosses a field slab of finite width. The family was reached through a four-iteration pilot in which every failure became a design constraint, and the pilot's central discovery is that the family has a measured critical manifold in its parameter plane. Above the manifold a deterministic crossing kick reverses every trajectory. Below it reversal is a rare fluctuation, and a thermal-free probe measures an effective gap that organizes the suppression.

Two sealed runs then interrogated that structure. The first asked whether the effective-gap axis predicts held-out cells to binomial precision and whether the Schwinger-shaped axis fails by a fixed factor. Its verdict was neither-axis. The effective-gap model organized the data well and beat the alternative decisively, but its residuals exceeded binomial scatter by far, so the sealed bars failed. The bars had tested a claim nobody made, that a single-axis phenomenological law holds to counting precision at fifty thousand trajectories per cell. The discipline caught its own designer, and the verdict stands in the record. Before any second attempt, the project owner imposed a standing rule. A second preregistration had to either model the two-variable structure of the exponent explicitly or set its bars against model error, and it had to declare which in writing first. The second sealed run took the second branch, on grids fully disjoint from everything measured before, and passed both bars with room.

The result is a disciplined negative. In this family the reversal exponent is set by the measure of the initial ensemble, which is exactly the outcome the campaign's generic negative control anticipated, and the Schwinger-

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shaped axis fails to compete. Section II defines the family and its measured structure. Sections III and IV report the two governed runs. Section V states what is and is not established, separated by claim type. Section VI records the evidence infrastructure.

II. THE FAMILY AND ITS MEASURED STRUCTURE

A. Definition

The Sauter-slab energy-transfer family evolves an event coordinate t and a hidden oscillator coordinate u under the Hamiltonian

$$H = \frac{1}{2}p_t^2 + \frac{1}{2}p_u^2 + \frac{1}{2}\omega_u^2 u^2 + \frac{\lambda}{4}u^4 + gEL \tanh(t/L) u, \quad (1)$$

with $\omega_t = 0$, so the time coordinate is free except for the coupling. The force on the time momentum is $-gEu \operatorname{sech}^2(t/L)$, a field localized in a slab of width L , so the field does bounded work on the event. The committed parameters are $(\omega_u, \lambda, g, T, L) = (1.2, 0.1, 0.25, 1, 3)$. The event enters from $t(0) = -4L$ with momentum gap $p_t(0) = P > 0$. The hidden oscillator is initialized thermally at temperature T about the Newton-solved shifted equilibrium of its effective well under the saturated tilt. Integration is velocity Verlet at $dt = 10^{-3}$ for forty units of internal time, and the scheme is symplectic here because every force depends on position only [7]. Reversal means $p_t \leq 0$ at any step. The family contains no exponential in its inputs, no field-dependent sampling, and no stopping rule correlated with reversal, in compliance with the anti-circularity contract.

B. Four iterations, each failure a constraint

The pilot reached this definition in four steps, and the record preserves all of them.

First, the original coupling $gEtu$ is unbounded in t , so secular growth let the field do unlimited work and the relative energy drift reached 0.23 at the strongest field. The slab replaced it, which is why the family conserves energy to 6×10^{-7} everywhere below.

Second, the slab version initialized the oscillator thermally about $u = 0$ while the saturated tilt already displaces the equilibrium at $t(0)$. Every ensemble member then carried the same coherent transient, whose deterministic momentum transfer reversed every trajectory, a measured fraction of 1.0. Nothing was a rare event. The fix initializes the ensemble about the shifted equilibrium.

Third, with the transient removed, a deterministic momentum transfer remained during the slab crossing, and it feeds back on itself, since a slowing event spends longer in the slab and absorbs more kick. The family therefore has a critical-field manifold $E_{\text{crit}}(P)$ separating deterministic reversal from fluctuation-driven reversal. The pilot turned

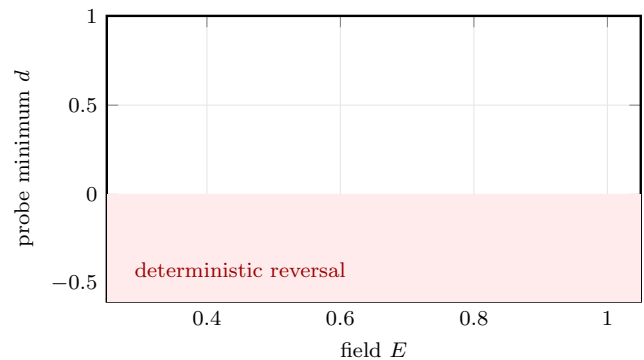


FIG. 1. The measured critical manifold of the pilot. Each point is the thermal-free probe’s deterministic minimum d of the time momentum for one pilot cell, plotted against the field for the three pilot gaps. The shaded region marks deterministic reversal, where the probe itself crosses zero and every trajectory reverses without fluctuations. The circled point is the one pilot cell in that regime. For $d > 0$ the probe minimum is the effective gap that organizes the rare-event suppression. Data are the probe entries of the committed pilot record.

this from a nuisance into an instrument. A thermal-free probe trajectory, started at the shifted equilibrium with no oscillator momentum, measures the deterministic minimum $d(P, E)$ of the time momentum in each cell and classifies its regime. Cells with $d \leq 0$ reverse deterministically and are excluded from every rare-event fit. For $d > 0$ the probe minimum is the family’s effective gap, the momentum the fluctuations must supply.

Fourth, the sub-critical transition is razor sharp. At gap $P = 0.5$ the measured fraction moved from 2.8×10^{-4} to 0.91 between fields 0.65 and 0.80, three decades of rate in a fifteen percent field step, and only two of the eighteen pilot cells landed in the measurable band. Fixed rectangular grids mostly buy bounds in this family. Both governed runs therefore place their cells adaptively, by bisecting the field until the probe hits declared targets in d/E , which uses no ensemble and no rate information.

C. Pilot suppression structure

The two measurable pilot cells admit one diagnostic, stated here with its caveat attached. Their suppressions satisfy $-\log f \propto (d/E)^2$ to two percent, a measured ratio of 2.03 against a predicted 1.99, while the Schwinger-shaped axis d^2/E misses by fifty-seven percent and the bare-gap axis $(P/E)^2$ by forty percent. Two points prove nothing, and this diagnostic is post hoc and exploratory. What it did was point the sealed design at the effective-gap Gaussian axis, which is the form a measure tail takes when reversal requires the thermal ensemble to supply momentum d against fluctuations whose scale is set by E , the classical escape-rate structure of Kramers [5]. The campaign’s generic negative control had already found

that fold statistics in a bounded Hamiltonian family are set rigidly by the dynamics and the initial-condition measure [8], so a measure-tail exponent was the expected shape.

III. THE FIRST SEALED RUN

A. Design

PREREG-PF4-001 declared the claim that the fluctuation-regime suppression is organized by the effective-gap Gaussian axis and not by the Schwinger-shaped axis. The two models are

$$\text{model G} \quad -\log f = \alpha + \beta (d/E)^2, \quad (2)$$

$$\text{model S} \quad -\log f = \alpha' + \beta' d^2/E, \quad (3)$$

fit by weighted least squares with binomial weights $\text{var}(-\log f) = (1-f)/(fN)$ on twelve training cells and scored on eight held-out cells. Training gaps were $P \in \{0.5, 0.7, 0.9\}$ and held-out gaps $P \in \{0.6, 0.8\}$, disjoint in P , with four cells per gap placed by bisection at probe targets $d/E \in \{0.20, 0.28, 0.36, 0.44\}$ and 5×10^4 trajectories per cell. The sealed bars were $\text{MSE}_G \leq 4$ in binomial units and $\text{MSE}_S \geq 4 \text{MSE}_G$. The claim would be refuted if model S matched or beat model G held out. Any other outcome was to be reported as neither-axis with both residual sets.

B. Outcome, verdict preserved

The instruments worked. All twenty probe-placed cells landed in the measurable band, so the adaptive manifest closed the pilot's grid problem. Drift was uniform at 6×10^{-7} . Both prescription controls agreed exactly, with $z = 0.00$ under an independent Runge-Kutta integration and under step halving. The fitted effective-gap slope was $\beta = 41.2$, confirming the pilot's two-point prediction of 41.6, and model G beat model S by a factor of 3.28 on held-out weighted error.

The sealed bars were still not met. The verdict was neither-axis, because $\text{MSE}_G = 81.2$ against a bar of 4, and model S, though clearly worse, was not four times worse. At 5×10^4 -trajectory precision the residual dependence of the suppression on the gap at fixed d/E is many binomial standard deviations. Stated plainly, the sealed bars tested a claim nobody made, that a single-axis phenomenological law holds to binomial precision. The effective-gap axis is a few-percent-accurate organizer of this family and not a law, and bars written in counting units could only discover that fact by failing. The discipline caught its own designer. The verdict is preserved in the record exactly as returned, and nothing in the second run modifies it.

IV. THE CORRECTION RULE AND THE SECOND SEALED RUN

A. The owner-imposed rule

Before any second attempt existed, the design document acquired a standing requirement. A second sealed attempt on this family must either model the two-variable structure explicitly, an exponent depending on more than one combination of P and E , or set its bars against model error rather than binomial error, and the preregistration must declare which of the two it does before any governed run executes. The rule was declared in writing before PREREG-PF4-002 was drafted, and the second preregistration opens by taking the second branch. Its bars are ratios to the model's own training inadequacy, so the claim under test is the one actually made, that the effective-gap axis organizes and generalizes while the Schwinger-shaped axis does not.

B. Design

PREREG-PF4-002 kept the family, the estimator, and the control rules of the first run unchanged and changed only the grids and the bar arithmetic. Training gaps were $P \in \{0.55, 0.75, 0.95\}$ and held-out gaps $P \in \{0.65, 0.85\}$, disjoint from the pilot grid and from the first governed run. Probe targets were $d/E \in \{0.22, 0.30, 0.38, 0.46\}$, disjoint from the first run's targets. The bars were that model G's held-out weighted error be at most twice its training weighted error, and that model S's held-out weighted error be at least twice model G's.

C. Outcome

Both bars passed with room. The generalization ratio was 0.38, so model G predicted the gaps it had never seen better than it fit the gaps it trained on. The Schwinger-shaped model was 3.14 times worse on the same held-out cells. Ten of twelve training cells and seven of eight held-out cells were usable under the sealed count and saturation rules, above the declared minimum. Drift was uniform at 6×10^{-7} . The verdict is a pass, and the claim carries the label demonstrated-in-model.

Both prescription controls again agreed exactly, with $z = 0.00$ under Runge-Kutta and under step halving, as they had in the first run. This exactness is an observation in its own right. The control cells sit at fractions near 0.01 and 0.1, where identical fractions across three prescriptions mean the integrators agree member for member on which of 5×10^4 trajectories reverse, not merely on the ensemble rate. At this step size the family's reversal classification is robust per trajectory, which is the strongest form of the closure clause's agreement requirement and the opposite of the prescription-borne threshold the clause exists to catch.

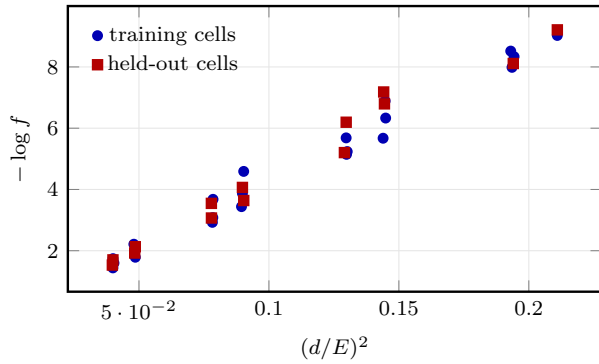


FIG. 2. Suppression against the effective-gap Gaussian axis for every usable cell of both governed runs, training and held-out marked separately, with the second run’s sealed G-model line overlaid. The axis organizes thirty-six cells spanning more than three decades of rate across six trained and four held-out gap values. The visible scatter about the line is real structure, many binomial standard deviations at this ensemble size, which is why the first run’s binomial-unit bars failed and the second run’s bars are ratios to model error. Data are the cell entries of the two committed run records.

V. WHAT THE RESULT MEANS AND DOES NOT MEAN

A. Claim register

The campaign separates claim types, and this paper’s claims sort as follows.

Theorem-level. The fold kinematics behind apparent pair events, the exact conservation argument that makes the static-potential family an analytic null with no dynamical exponential, and the frozen-kick result that impulsively closed interactions cannot produce continuous reversal [9] are proved statements, not measurements.

Sealed and measured. The neither-axis verdict of PREREG-PF4-001 and the passing verdict of PREREG-PF4-002 are governed outcomes under sealed bars, with the second carrying the label demonstrated-in-model. Together they establish that in the declared Sauter-slab family the reversal suppression is organized by the effective-gap Gaussian axis within its own accuracy class, generalizing across gaps it never trained on, and that the Schwinger-shaped axis fails to compete.

Exploratory. The pilot sweep, the critical-manifold map, the two-point diagnostic, and the slope confirmation $\beta = 41.2$ against the predicted 41.6 are unsealed or secondary observations. They shaped the design and none of them bears the claim.

Open. Whether any constructible hidden-dynamics family has a reversal exponent that is not a measure tail is not addressed by any run reported here.

B. Meaning

The exponent in this family is measure-set. Reversal happens when the thermal ensemble supplies the effective gap, so the suppression is the Gaussian tail of the declared initial measure, evaluated in a variable the dynamics itself defines through the probe. This is exactly the shape the campaign’s generic negative control anticipated for bounded local Hamiltonian families, now established under seal rather than observed in passing. The Schwinger form, which differs from the measure-tail form by one power of the field in the exponent, had a fair preregistered chance on held-out data twice and failed to compete both times.

C. Limits

The verdict binds one declared family at its declared parameters. It licenses no statement about other hidden-dynamics families, none about physical pair production, and none about quantum electrodynamics, whose exponent is derived structure, not a fitted axis [3, 4]. The open summit of the program is now sharply posed. Either some constructible family has a reversal exponent that is not the tail of its own initial measure, or the Schwinger exponential is unreachable by this route. Any claimed positive would owe its audience what this paper’s negative paid, continuous integration under the closure clause, prescription-pair and refinement controls at every fitted point, the planted-artifact check, disjoint held-out grids, and bars declared against the claim actually made.

VI. METHODS AND REPRODUCIBILITY

The campaign’s instruments were sealed under tolerance freezes before any claim-bearing run, and the seal ledger now carries six entries, two instrument freezes and four preregistrations, each a triple of registration identifier, sealing commit, and content hash. Evidence records are append-only JSON with SHA-256 hashes over their own content, and the three records behind this paper are `results/pf4-pilot.json`, `results/prereg-pf4-001.json`, and `results/prereg-pf4-002.json`. Every number quoted here appears verbatim in one of those records or in the sealed design documents, and both figures are generated by a committed script whose only inputs are the committed records.

The closure clause was active throughout. All dynamics were integrated continuously with velocity Verlet at the declared step, the prescription was preregistered, and the prescription-pair and refinement controls ran on cells verified measurable in advance, closing a pilot lesson in which a control cell with zero counts had certified nothing. The planted Cayley-trap control, clause C5, was deferred in both preregistrations with the recorded justification that

these runs compare axes within one declared continuous prescription rather than asserting a rate value against an external target, while C3 and C4 remained active. The deferral is not a waiver. Any future preregistration that

claims a rate law against a physical target must implement the full trap, and a pipeline that fails to flag the planted artifact voids its own run.

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